Second Partial

Fco. Vazquez

Tec de Monterrey

Class Activity

Show that the critical point of $h(t) = -\frac{g}{2}t^2 + v_o t + h_o$ is $\left(\dfrac{v_o}{g}, \dfrac{v_o^2}{2g} + h_o\right)$

$$\frac{\mathrm{d}}{\mathrm{d}x} u(x)v(x) = u'(x)v(x) + u(x)v'(x)$$
$$\frac{\mathrm{d}}{\mathrm{d}x} \frac{u(x)}{v(x)} = \frac{1}{v^2(x)} \left[u'(x)v(x) - u(x)v'(x)\right]$$
$$\frac{\mathrm{d}}{\mathrm{d}x} u(v(x)) = u'(v(x))v'(x)$$

$$a^n \cdot a^m = a^{n+m}, \qquad \log_n[a] + \log_n[b] = \log_n[a \cdot b]$$
$$(a^n)^m = a^{n \cdot m}, \qquad a \log_n[b] = \log_n[b^a]$$
$$\frac{1}{a} = a^{-1}, \qquad \log_n[a] - \log_n[b] = \log_n\left[\frac{a}{b}\right]$$
$$a^{\log_a[x]} = x = \log_a[a^x], \qquad \log_n[x] = \frac{\log_m[x]}{\log_m[n]}$$